***Proof.*** By Theorem 6.1.3(c), the derivative $(FG)'$ exists on $[a, b]$ and

$$(FG)' = F'G + FG' = fG + Fg.$$

Since $F$, $G$ are continuous and $f$, $g$ belong to $\mathcal{R}[a, b]$, the Product Theorem 7.3.16 implies that $fG$ and $Fg$ are integrable. Therefore the Fundamental Theorem 7.3.1 implies that

$$FG\Big|_a^b = \int_a^b (FG)' = \int_a^b fG + \int_a^b Fg,$$

from which (7) follows. Q.E.D.

A special, but useful, case of this theorem is when $f$ and $g$ are continuous on $[a, b]$ and $F$, $G$ are their indefinite integrals $F(x) := \int_a^x f$ and $G(x) := \int_a^x g$.

We close this section with a version of Taylor's Theorem for the Riemann Integral.

**7.3.18 Taylor's Theorem with the Remainder** *Suppose that $f', \ldots, f^{(n)}, f^{(n+1)}$ exist on $[a, b]$ and that $f^{(n+1)} \in \mathcal{R}[a, b]$. Then we have*

$$f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \cdots + \frac{f^{(n)}(a)}{n!}(b-a)^n + R_n, \tag{8}$$

*where the remainder is given by*

$$R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\cdot(b-t)^n dt. \tag{9}$$

***Proof.*** Apply Integration by Parts to equation (9), with $F(t) := f^{(n)}(t)$ and $G(t) := (b-t)^n/n!$, so that $g(t) = -(b-t)^{n-1}/(n-1)!$, to get

$$\begin{aligned} R_n &= \frac{1}{n!}f^{(n)}(t)\cdot(b-t)^n\Big|_{t=a}^{t=b} + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)\cdot(b-a)^{n-1}dt \\ &= -\frac{f^{(n)}(a)}{n!}\cdot(b-a)^n + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)\cdot(b-t)^{n-1}dt. \end{aligned}$$

If we continue to integrate by parts in this way, we obtain (8). Q.E.D.

## Exercises for Section 7.3

1. Extend the proof of the Fundamental Theorem 7.3.1 to the case of an arbitrary finite set $E$.
2. If $n \in \mathbb{N}$ and $H_n(x) := x^{n+1}/(n+1)$ for $x \in [a, b]$, show that the Fundamental Theorem 7.3.1 implies that $\int_a^b x^n dx = \left(b^{n+1} - a^{n+1}\right)/(n+1)$. What is the set $E$ here?
3. If $g(x) := x$ for $|x| \geq 1$ and $g(x) := -x$ for $|x| < 1$ and if $G(x) := \frac{1}{2}\left|x^2 - 1\right|$, show that $\int_{-2}^3 g(x)dx = G(3) - G(-2) = 5/2$.
4. Let $B(x) := -\frac{1}{2}x^2$ for $x < 0$ and $B(x) := \frac{1}{2}x^2$ for $x \geq 0$. Show that $\int_a^b |x|dx = B(b) - B(a)$.
5. Let $f : [a, b] \to \mathbb{R}$ and let $C \in \mathbb{R}$.
   (a) If $\Phi : [a, b] \to \mathbb{R}$ is an antiderivative of $f$ on $[a, b]$, show that $\Phi_C(x) := \Phi(x) + C$ is also an antiderivative of $f$ on $[a, b]$.
   (b) If $\Phi_1$ and $\Phi_2$ are antiderivatives of $f$ on $[a, b]$, show that $\Phi_1 - \Phi_2$ is a constant function on $[a, b]$.

6. If $f \in \mathcal{R}[a,b]$ and if $c \in [a,b]$, the function defined by $F_c(z) := \int_c^z f$ for $z \in [a,b]$ is called the **indefinite integral** of $f$ with **basepoint** $c$. Find a relation between $F_a$ and $F_c$.

7. We have seen in Example 7.1.7 that Thomae's function is in $\mathcal{R}[0,1]$ with integral equal to 0. Can the Fundamental Theorem 7.3.1 be used to obtain this conclusion? Explain your answer.

8. Let $F(x)$ be defined for $x \geq 0$ by $F(x) := (n-1)x - (n-1)n/2$ for $x \in [n-1, n)$, $n \in \mathbb{N}$. Show that $F$ is continuous and evaluate $F'(x)$ at points where this derivative exists. Use this result to evaluate $\int_a^b [\![x]\!]dx$ for $0 \leq a < b$, where $[\![x]\!]$ denotes the greatest integer in $x$, as defined in Exercise 5.1.4.

9. Let $f \in \mathcal{R}[a,b]$ and define $F(x) := \int_a^x f$ for $x \in [a,b]$.
   (a) Evaluate $G(x) := \int_c^x f$ in terms of $F$, where $c \in [a,b]$.
   (b) Evaluate $H(x) := \int_x^b f$ in terms of $F$.
   (c) Evaluate $S(x) := \int_x^{\sin x} f$ in terms of $F$.

10. Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a, b]$ and let $v : [c,d] \to \mathbb{R}$ be differentiable on $[c, d]$ with $v([c,d]) \subseteq [a,b]$. If we define $G(x) := \int_a^{v(x)} f$, show that $G'(x) = f(v(x)) \cdot v'(x)$ for all $x \in [c,d]$.

11. Find $F'(x)$ when $F$ is defined on [0, 1] by:
    (a) $F(x) := \int_0^{x^2} (1+t^3)^{-1} dt$. (b) $F(x) := \int_{x^2}^x \sqrt{1+t^2}dt$.

12. Let $f : [0, 3] \to \mathbb{R}$ be defined by $f(x) := x$ for $0 \leq x < 1$, $f(x) := 1$ for $1 \leq x < 2$ and $f(x) := x$ for $2 \leq x \leq 3$. Obtain formulas for $F(x) := \int_0^x f$ and sketch the graphs of $f$ and $F$. Where is $F$ differentiable? Evaluate $F'(x)$ at all such points.

13. The function $g$ is defined on [0, 3] by $g(x) := -1$ if $0 \leq x < 2$ and $g(x) := 1$ if $2 \leq x \leq 3$. Find the indefinite integral $G(x) = \int_0^x g$ for $0 \leq x \leq 3$, and sketch the graphs of $g$ and $G$. Does $G'(x) = g(x)$ for all $x$ in [0, 3]?

14. Show there does not exist a continuously differentiable function $f$ on [0, 2] such that $f(0) = -1$, $f(2) = 4$, and $f'(x) \leq 2$ for $0 \leq x \leq 2$. (Apply the Fundamental Theorem.)

15. If $f : \mathbb{R} \to \mathbb{R}$ is continuous and $c > 0$, define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) := \int_{x-c}^{x+c} f(t)dt$. Show that $g$ is differentiable on $\mathbb{R}$ and find $g'(x)$.

16. If $f : [0, 1] \to \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0, 1]$, show that $f(x) = 0$ for all $x \in [0, 1]$.

17. Use the following argument to prove the Substitution Theorem 7.3.8. Define $F(u) := \int_{\varphi(\alpha)}^u f(x)dx$ for $u \in I$, and $H(t) := F(\varphi(t))$ for $t \in J$. Show that $H'(t) = f(\varphi(t))\varphi'(t)$ for $t \in J$ and that

$$\int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)dx = F(\varphi(\beta)) = H(\beta) = \int_\alpha^\beta f(\varphi(t))\varphi'(t)dt.$$

18. Use the Substitution Theorem 7.3.8 to evaluate the following integrals.
    (a) $\int_0^1 t\sqrt{1+t^2}dt$, (b) $\int_0^2 t^2(1+t^3)^{-1/2}dt = 4/3$,
    (c) $\int_1^4 \frac{\sqrt{1+\sqrt{t}}}{\sqrt{t}}dt$, (d) $\int_1^4 \frac{\cos\sqrt{t}}{\sqrt{t}}dt = 2(\sin 2 - \sin 1)$.

19. Explain why Theorem 7.3.8 and/or Exercise 7.3.17 cannot be applied to evaluate the following integrals, using the indicated substitution.
    (a) $\int_0^4 \frac{\sqrt{t}dt}{1+\sqrt{t}}$ $\varphi(t) = \sqrt{t}$, (b) $\int_0^4 \frac{\cos\sqrt{t}\,dt}{\sqrt{t}}$ $\varphi(t) = \sqrt{t}$,
    (c) $\int_{-1}^1 \sqrt{1+2|t|}dt$ $\varphi(t) = |t|$, (d) $\int_0^1 \frac{dt}{\sqrt{1-t^2}}$ $\varphi(t) = \text{Arcsin } t$.

20. (a) If $Z_1$ and $Z_2$ are null sets, show that $Z_1 \cup Z_2$ is a null set.
    (b) More generally, if $Z_n$ is a null set for each $n \in \mathbb{N}$, show that $\bigcup_{n=1}^{\infty} Z_n$ is a null set. [*Hint:* Given $\varepsilon > 0$ and $n \in \mathbb{N}$, let $\{J_k^n : k \in \mathbb{N}\}$ be a countable collection of open intervals whose union contains $Z_n$ and the sum of whose lengths is $\leq \varepsilon/2^n$. Now consider the countable collection $\{J_k^n : n, k \in \mathbb{N}\}$.]

21. Let $f, g \in \mathcal{R}[a, b]$.
    (a) If $t \in \mathbb{R}$, show that $\int_a^b (tf \pm g)^2 \geq 0$.
    (b) Use (a) to show that $2\left|\int_a^b fg\right| \leq t \int_a^b f^2 + (1/t) \int_a^b g^2$ for $t > 0$.
    (c) If $\int_a^b f^2 = 0$, show that $\int_a^b fg = 0$.
    (d) Now prove that $\left|\int_a^b fg\right|^2 \leq \left(\int_a^b |fg|\right)^2 \leq \left(\int_a^b f^2\right) \cdot \left(\int_a^b g^2\right)$. This inequality is called the **Cauchy-Bunyakovsky-Schwarz Inequality** (or simply the **Schwarz Inequality**).

22. Let $h : [0, 1] \to \mathbb{R}$ be Thomae's function and let sgn be the signum function. Show that the composite function $\text{sgn} \circ h$ is not Riemann integrable on [0, 1].

## Section 7.4 The Darboux Integral

An alternative approach to the integral is due to the French mathematician Gaston Darboux (1842–1917). Darboux had translated Riemann's work on integration into French for publication in a French journal and inspired by a remark of Riemann, he developed a treatment of the integral in terms of upper and lower integrals that was published in 1875. Approximating sums in this approach are obtained from partitions using the infima and suprema of function values on subintervals, which need not be attained as function values and thus the sums need not be Riemann sums.

This approach is technically simpler in the sense that it avoids the complications of working with infinitely many possible choices of tags. But working with infima and suprema also has its complications, such as lack of additivity of these quantities. Moreover, the reliance on the order properties of the real numbers causes difficulties in extending the Darboux integral to higher dimensions, and, more importantly, impedes generalization to more abstract surfaces such as manifolds. Also, the powerful Henstock-Kurzweil approach to integration presented in Chapter 10, which includes the Lebesgue integral, is based on the Riemann definition as given in Section 7.1.

In this section we introduce the upper and lower integrals of a bounded function on an interval, and define a function to be Darboux integrable if these two quantities are equal. We then look at examples and establish a Cauchy-like integrability criterion for the Darboux integral. We conclude the section by proving that the Riemann and Darboux approaches to the integral are in fact equivalent, that is, a function on a closed, bounded interval is Riemann integrable if and only if it is Darboux integrable. Later topics in the book do not depend on the Darboux definition of integral so that this section can be regarded as optional.

### Upper and Lower Sums

Let $f : I \to \mathbb{R}$ be a bounded function on $I = [a, b]$ and let $\mathcal{P} = (x_0, x_1, \ldots, x_n)$ be a partition of $I$. For $k = 1, 2, \ldots, n$ we let

$$m_k := \inf\{f(x) : x \in [x_{k-1}, x_k]\}, \qquad M_k := \sup\{f(x) : x \in [x_{k-1}, x_k]\}.$$